

Carga de librerias

```
In [4]: #Hacer Llamadas http a paginas de La red
!pip install requests
#Modulo para Las instancias del problema del TSP
!pip install tsplib95

Requirement already satisfied: requests in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (2.32.5)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from requests) (2026.2.25)
Requirement already satisfied: tsplib95 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (0.7.1)
Requirement already satisfied: Click>=6.0 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from tsplib95) (8.3.2)
Requirement already satisfied: Deprecated~=1.2.9 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from tsplib95) (1.2.18)
Collecting networkx~=2.1 (from tsplib95)
  Using cached networkx-2.8.8-py3-none-any.whl.metadata (5.1 kB)
Requirement already satisfied: tabulate~=0.8.7 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from tsplib95) (0.8.10)
Requirement already satisfied: wrapt<2,>=1.10 in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from Deprecated~=1.2.9->tsplib95) (1.17.3)
Requirement already satisfied: colorama in c:\users\carlos\anaconda3\envs\ia_env\lib\site-packages (from Click>=6.0->tsplib95) (0.4.6)
Using cached networkx-2.8.8-py3-none-any.whl (2.0 MB)
Installing collected packages: networkx
  Attempting uninstall: networkx
    Found existing installation: networkx 3.6.1
    Uninstalling networkx-3.6.1:
      Successfully uninstalled networkx-3.6.1
Successfully installed networkx-2.8.8

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
scikit-image 0.26.0 requires networkx>=3.0, but you have networkx 2.8.8 which is incompatible.
```

Carga de los datos del problema

```
In [8]: import urllib.request #Hacer Llamadas http a paginas de La red
import tsplib95 #Modulo para Las instancias del problema del TSP
import math #Modulo de funciones matematicas. Se usa para exp
import random #Para generar valores aleatorios

#http://elib.zib.de/pub/mp-testdata/tsp/tsplib/
#Documentacion :
# http://comopt.ifl.uni-heidelberg.de/software/TSPLIB95/tsp95.pdf
# https://tsplib95.readthedocs.io/en/stable/pages/usage.html
# https://tsplib95.readthedocs.io/en/v0.6.1/modules.html
# https://pypi.org/project/tsplib95/

#Descargamos el fichero de datos(Matriz de distancias)

#En este caso se descarga el dataset y se carga de una ubicación local
file = "swiss42.tsp" ;
urllib.request.urlretrieve("http://comopt.ifl.uni-heidelberg.de/software/TSPLIB95/tsp/swiss42.tsp.gz", file + '.gz')
!gzip -d swiss42.tsp.gz #Descomprimir el fichero de datos

#Coordendas 51-city problem (Christofides/Eilon)
#file = "eil51.tsp" ; urllib.request.urlretrieve("http://comopt.ifl.uni-heidelberg.de/software/TSPLIB95/tsp/eil51.tsp.gz", file)
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#Coordenadas - 48 capitals of the US (Padberg/Rinaldi)
#file = "att48.tsp" ; urllib.request.urlretrieve("http://comopt.ifi.uni-heidelberg.de/software/TSPLIB95/tsp/att48.tsp.gz", file)
```

```
In [9]: #Carga de datos y generación de objeto problem
#####
problem = tsplib95.load(file)

#Nodos
Nodos = list(problem.get_nodes())

#Aristas
Aristas = list(problem.get_edges())
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(22, 33),
(22, 34),
(22, 35),

(22, 36),
(22, 37),
(22, 38),
(22, 39),
(22, 40),
(22, 41),
(23, 0),
(23, 1),
(23, 2),
(23, 3),
(23, 4),
(23, 5),
(23, 6),
(23, 7),
(23, 8),
(23, 9),
(23, 10),
(23, 11),
(23, 12),
(23, 13),
(23, 14),
(23, 15),
(23, 16),
(23, 17),
(23, 18),
(23, 19),
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(23, 22),
(23, 23),
(23, 24),
(23, 25),
(23, 26),
(23, 27),
(23, 28),
(23, 29),
(23, 30),
(23, 31),
(23, 32),
(23, 33),
...]

```
NOMBRE: swiss42
TIPO: TSP
COMENTARIO: 42 Staedte Schweiz (Fricker)
DIMENSION: 42
EDGE_WEIGHT_TYPE: EXPLICIT
EDGE_WEIGHT_FORMAT: FULL_MATRIX
EDGE_WEIGHT_SECTION
0 15 30 23 32 55 33 37 92 114 92 110 96 90 74 76 82 72 78 82 159 122 131 206 112 57 28 43 70 1
15 0 34 23 27 40 19 32 93 117 88 100 87 75 63 67 71 69 62 63 96 164 132 131 212 106 44 33 5
30 34 0 11 18 57 36 65 62 84 64 89 76 93 95 100 104 98 57 88 99 130 100 101 179 86 51 4 18 4
23 23 11 0 11 48 26 54 70 94 69 75 75 84 84 89 92 89 54 78 99 141 111 109 89 89 11 11 11 54
32 27 18 11 0 40 20 58 67 92 61 78 65 76 83 89 91 95 43 72 110 141 116 105 190 81 34 19 35 1
55 40 57 48 40 0 23 55 96 123 78 75 36 36 66 66 63 95 34 34 137 174 156 129 224 90 15 59 75
33 19 36 26 20 23 0 45 85 111 75 82 69 60 63 70 71 85 44 52 115 161 136 122 210 91 25 37 54
37 32 65 54 58 55 45 0 124 149 118 126 113 80 42 42 40 40 87 87 94 158 158 163 242 135 65 6
92 93 62 70 67 96 85 124 0 28 29 68 63 122 148 155 156 159 67 129 148 78 80 39 129 46 82 65
114 117 84 94 92 123 111 149 28 0 54 91 88 150 174 181 182 181 95 157 159 50 65 27 102 65 110
92 88 64 69 61 78 75 118 29 54 0 39 34 99 134 142 141 157 44 110 161 103 109 52 154 22 63 61
110 100 89 89 78 75 82 126 68 91 39 0 14 80 129 139 135 167 39 98 187 136 148 81 186 28 61 91
96 87 76 75 65 62 69 113 63 88 34 14 0 72 117 128 124 153 26 88 174 136 142 82 187 32 48 79
90 75 93 84 76 36 60 80 122 150 99 80 72 0 59 71 63 116 56 25 170 201 189 151 252 104 44 95
74 63 95 84 83 56 63 42 148 174 134 129 117 59 0 11 8 63 93 35 135 223 195 184 273 146 71 91
```

```
In [12]: #Probamos algunas funciones del objeto problem

#Distancia entre nodos
problem.get_weight(0, 1)

#Todas Las funciones
#Documentación: https://tsplib95.readthedocs.io/en/v0.6.1/modules.html

#dir(problem)
```

Out[12]: 15

Funcionas basicas

```
In [13]: #Funcionas basicas
#####

#Se genera una solucion aleatoria con comienzo en en el nodo 0
def crear_solucion(Nodos):
    solucion = [Nodos[0]]
    for n in Nodos[1:]:
        solucion = solucion + [random.choice(list(set(Nodos) - set({Nodos[0]}) - set(solucion)))]
    return solucion

#Devuelve La distancia entre dos nodos
def distancia(a,b, problem):
    return problem.get_weight(a,b)

#Devuelve La distancia total de una trayectoria/solucion
def distancia_total(solucion, problem):
    distancia_total = 0
    for i in range(len(solucion)-1):
        distancia_total += distancia(solucion[i] ,solucion[i+1] , problem)
    return distancia_total + distancia(solucion[len(solucion)-1] ,solucion[0], problem)

sol_temporal = crear_solucion(Nodos)
```

```
distancia_total(sol_temporal, problem), sol_temporal
```

Out[13]: (4634,
[0,
16,
30,
25,
14,
31,
20,
5,
24,
28,
27,
22,
2,
34,
37,
13,
36,
33,
26,
6,
35,
11,
8,
19,
23,
38,
12,
15,
17,
7,
9,
21,
1,
3,
39,
10,
4,
41,
18,
40,
29,
32])

BUSQUEDA ALEATORIA

```
In [15]: #####  
# BUSQUEDA ALEATORIA  
#####  
  
def busqueda_aleatoria(problem, N):  
    #N es el numero de iteraciones  
    Nodos = list(problem.get_nodes())  
  
    mejor_solucion = []
```

```
#mejor_distancia = 10e100                                #Inicializamos con un valor alto
mejor_distancia = float('inf')                            #Inicializamos con un valor alto

for i in range(N):                                       #Criterio de parada: repetir N veces pero podemos incluir otros
    solucion = crear_solucion(Nodos)                    #Genera una solucion aleatoria
    distancia = distancia_total(solucion, problem)      #Calcula el valor objetivo(distancia total)

    if distancia < mejor_distancia:                    #Compara con la mejor obtenida hasta ahora
        mejor_solucion = solucion
        mejor_distancia = distancia

print("Mejor solución:" , mejor_solucion)
print("Distancia      :" , mejor_distancia)
return mejor_solucion

#Busqueda aleatoria con 5000 iteraciones
solucion = busqueda_aleatoria(problem, 10000)
```

Mejor solución: [0, 23, 24, 41, 1, 7, 4, 14, 18, 10, 2, 29, 33, 11, 31, 36, 35, 37, 15, 6, 32, 34, 13, 12, 26, 20, 8, 40, 21, 39, 27, 19, 5, 38, 28, 30, 25, 22, 9, 16, 17, 3]
Distancia : 3723

BUSQUEDA LOCAL

```
In [17]: #####
# BUSQUEDA LOCAL
#####
def genera_vecina(solucion):
    #Generador de soluciones vecinas: 2-opt (intercambiar 2 nodos) Si hay N nodos se generan (N-1)x(N-2)/2 soluciones
    #Se puede modificar para aplicar otros generadores distintos que 2-opt
    #print(solucion)
    mejor_solucion = []
    mejor_distancia = 10e100
    for i in range(1,len(solucion)-1):                  #Recorremos todos Los nodos en bucle doble para evaluar todos Los intercambios 2-opt
        for j in range(i+1, len(solucion)):

            #Se genera una nueva solución intercambiando Los dos nodos i,j:
            # (usamos el operador + que para listas en python Las concatena) : ej.: [1,2] + [3] = [1,2,3]
            vecina = solucion[:i] + [solucion[j]] + solucion[i+1:j] + [solucion[i]] + solucion[j+1:]

            #Se evalua la nueva solución ...
            distancia_vecina = distancia_total(vecina, problem)

            #... para guardarla si mejora Las anteriores
            if distancia_vecina <= mejor_distancia:
                mejor_distancia = distancia_vecina
                mejor_solucion = vecina
    return mejor_solucion

#solucion = [1, 47, 13, 41, 40, 19, 42, 44, 37, 5, 22, 28, 3, 2, 29, 21, 50, 34, 30, 9, 16, 11, 38, 49, 10, 39, 33, 45, 15, 24, 43, 26, 31, 36, 35, 20, 8, 7, 23, 48, 27, 12, 17, 4, 18, 25, 14]
print("Distancia Solucion Incial:" , distancia_total(solucion, problem))

nueva_solucion = genera_vecina(solucion)
print("Distancia Mejor Solucion Local:", distancia_total(nueva_solucion, problem))
```


Distancia Solucion Incial: 3723
Distancia Mejor Solucion Local: 3274

```
In [18]: #Busqueda Local:
# - Sobre el operador de vecindad 2-opt(funcion genera_vecina)
# - Sin criterio de parada, se para cuando no es posible mejorar.
def busqueda_local(problem):
    mejor_solucion = []

    #Generar una solucion inicial de referencia(aleatoria)
    solucion_referencia = crear_solucion(Nodos)
    mejor_distancia = distancia_total(solucion_referencia, problem)

    iteracion=0          #Un contador para saber las iteraciones que hacemos
    while(1):
        iteracion +=1      #Incrementamos el contador
        #print('#',iteracion)

        #Obtenemos la mejor vecina ...
        vecina = genera_vecina(solucion_referencia)

        #... y la evaluamos para ver si mejoramos respecto a lo encontrado hasta el momento
        distancia_vecina = distancia_total(vecina, problem)

        #Si no mejoramos hay que terminar. Hemos llegado a un minimo local(según nuestro operador de vecindad 2-opt)
        if distancia_vecina < mejor_distancia:
            #mejor_solucion = copy.deepcopy(vecina)    #Con copia profunda. Las copias en python son por referencia
            mejor_solucion = vecina                  #Guarda la mejor solución encontrada
            mejor_distancia = distancia_vecina

        else:
            print("En la iteracion ", iteracion, ", la mejor solución encontrada es:" , mejor_solucion)
            print("Distancia      :" , mejor_distancia)
            return mejor_solucion

    solucion_referencia = vecina

sol = busqueda_local(problem )
```

En la iteracion 36 , la mejor solución encontrada es: [0, 27, 2, 28, 30, 29, 8, 11, 12, 18, 26, 7, 17, 31, 36, 35, 33, 20, 10, 25, 41, 23, 9, 4, 6, 37, 15, 16, 14, 19, 13, 5, 22, 39, 21, 40, 24, 38, 34, 32, 3, 1]
Distancia : 1844

SIMULATED ANNEALING

```
In [23]: #####
# SIMULATED ANNEALING
#####

#Generador de 1 solucion vecina 2-opt 100% aleatoria (intercambiar 2 nodos)
#Mejorable eligiendo otra forma de elegir una vecina.
def genera_vecina_aleatorio(solucion):

    #Se eligen dos nodos aleatoriamente
    i,j = sorted(random.sample( range(1,len(solucion)) , 2))

    #Devuelve una nueva solución pero intercambiando Los dos nodos elegidos al azar
```

```

    return solucion[:i] + [solucion[j]] + solucion[i+1:j] + [solucion[i]] + solucion[j+1:]

#Funcion de probabilidad para aceptar peores soluciones
def probabilidad(T,d):
    if random.random() < math.exp( -1*d / T) :
        return True
    else:
        return False

#Funcion de descenso de temperatura
def bajar_temperatura(T):
    return T*0.99

```

```

In [24]: def recocido_simulado(problem, TEMPERATURA ):
    #problem = datos del problema
    #T = Temperatura

    solucion_referencia = crear_solucion(Nodos)
    distancia_referencia = distancia_total(solucion_referencia, problem)

    mejor_solucion = []           #x* del pseudocodigo
    mejor_distancia = 10e100      #F* del pseudocodigo

    N=0
    while TEMPERATURA > .0001:
        N+=1
        #Genera una solución vecina
        vecina =genera_vecina_aleatorio(solucion_referencia)

        #Calcula su valor(distancia)
        distancia_vecina = distancia_total(vecina, problem)

        #Si es la mejor solución de todas se guarda(siempre!!!)
        if distancia_vecina < mejor_distancia:
            mejor_solucion = vecina
            mejor_distancia = distancia_vecina

        #Si la nueva vecina es mejor se cambia
        #Si es peor se cambia según una probabilidad que depende de T y delta(distancia_referencia - distancia_vecina)
        if distancia_vecina < distancia_referencia or probabilidad(TEMPERATURA, abs(distancia_referencia - distancia_vecina) ) :
            #solucion_referencia = copy.deepcopy(vecina)
            solucion_referencia = vecina
            distancia_referencia = distancia_vecina

        #Bajamos la temperatura
        TEMPERATURA = bajar_temperatura(TEMPERATURA)

    print("La mejor solución encontrada es " , end="")
    print(mejor_solucion)
    print("con una distancia total de " , end="")
    print(mejor_distancia)
    return mejor_solucion

sol = recocido_simulado(problem, 10000000)

```

La mejor solución encontrada es [0, 5, 13, 19, 6, 3, 2, 8, 9, 41, 10, 26, 14, 16, 15, 7, 1, 32, 30, 29, 28, 18, 12, 11, 25, 23, 21, 39, 22, 27, 4, 37, 36, 35, 38, 24, 40, 31, 17, 20, 33, 34]
con una distancia total de 2225

In []: